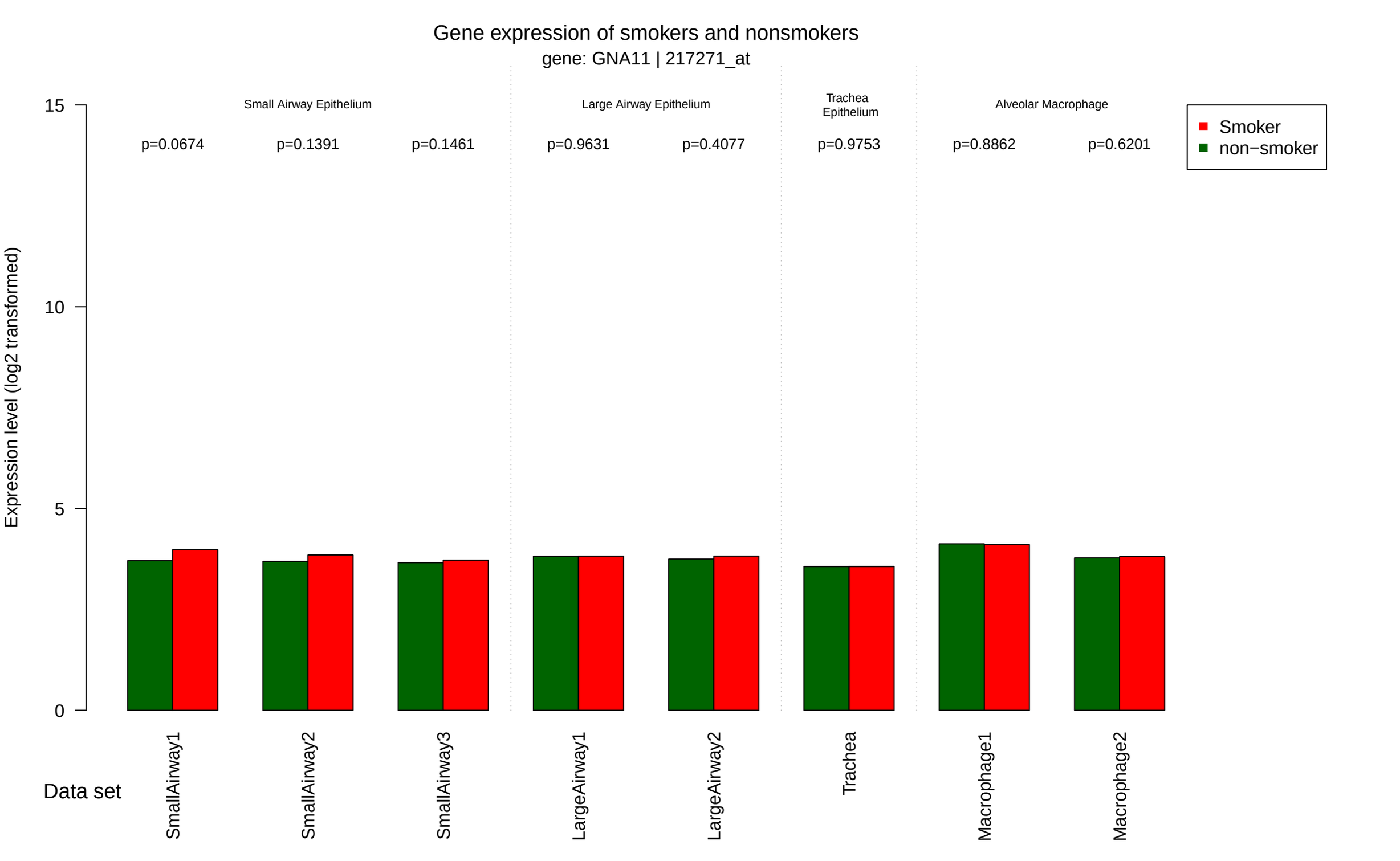
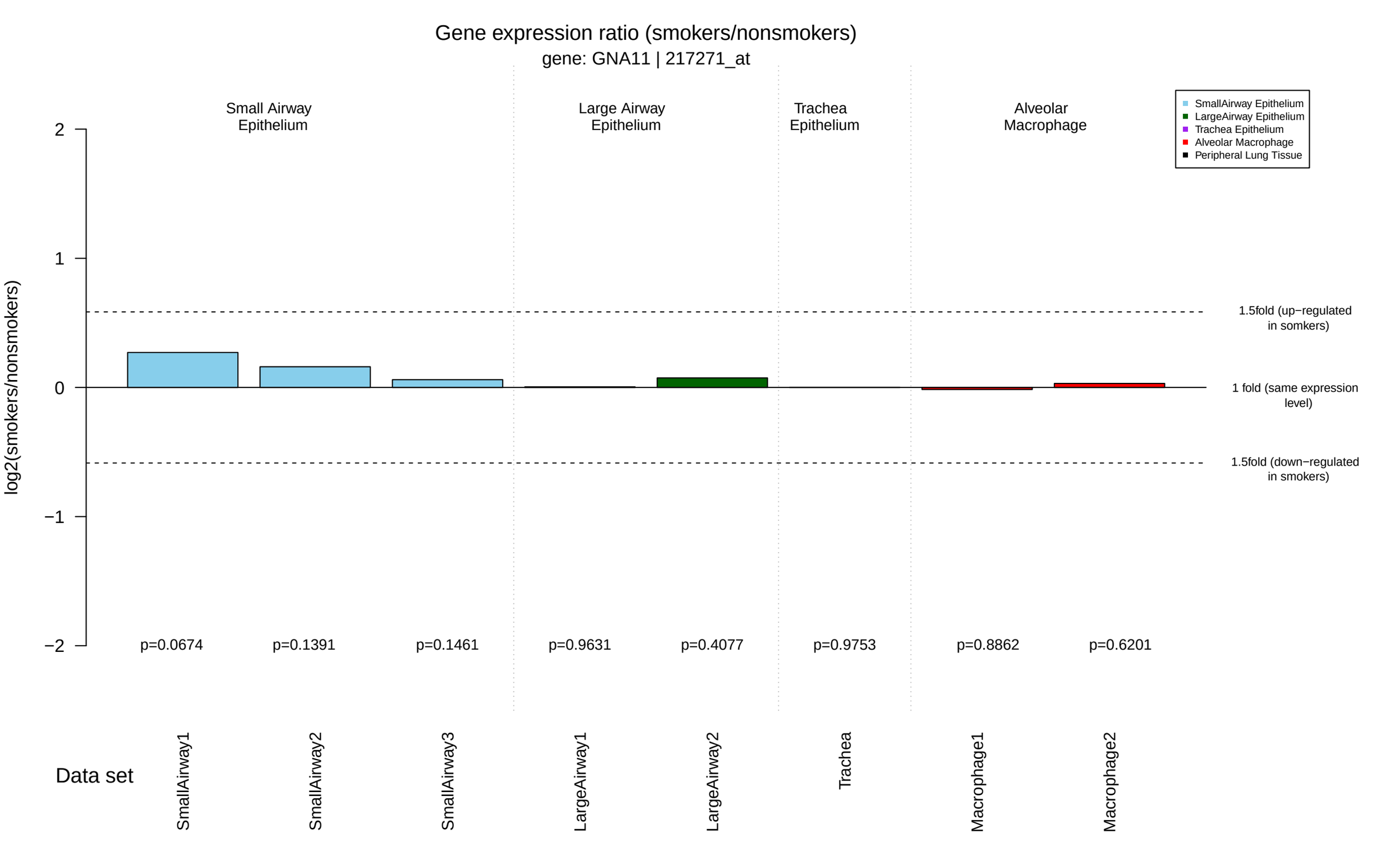
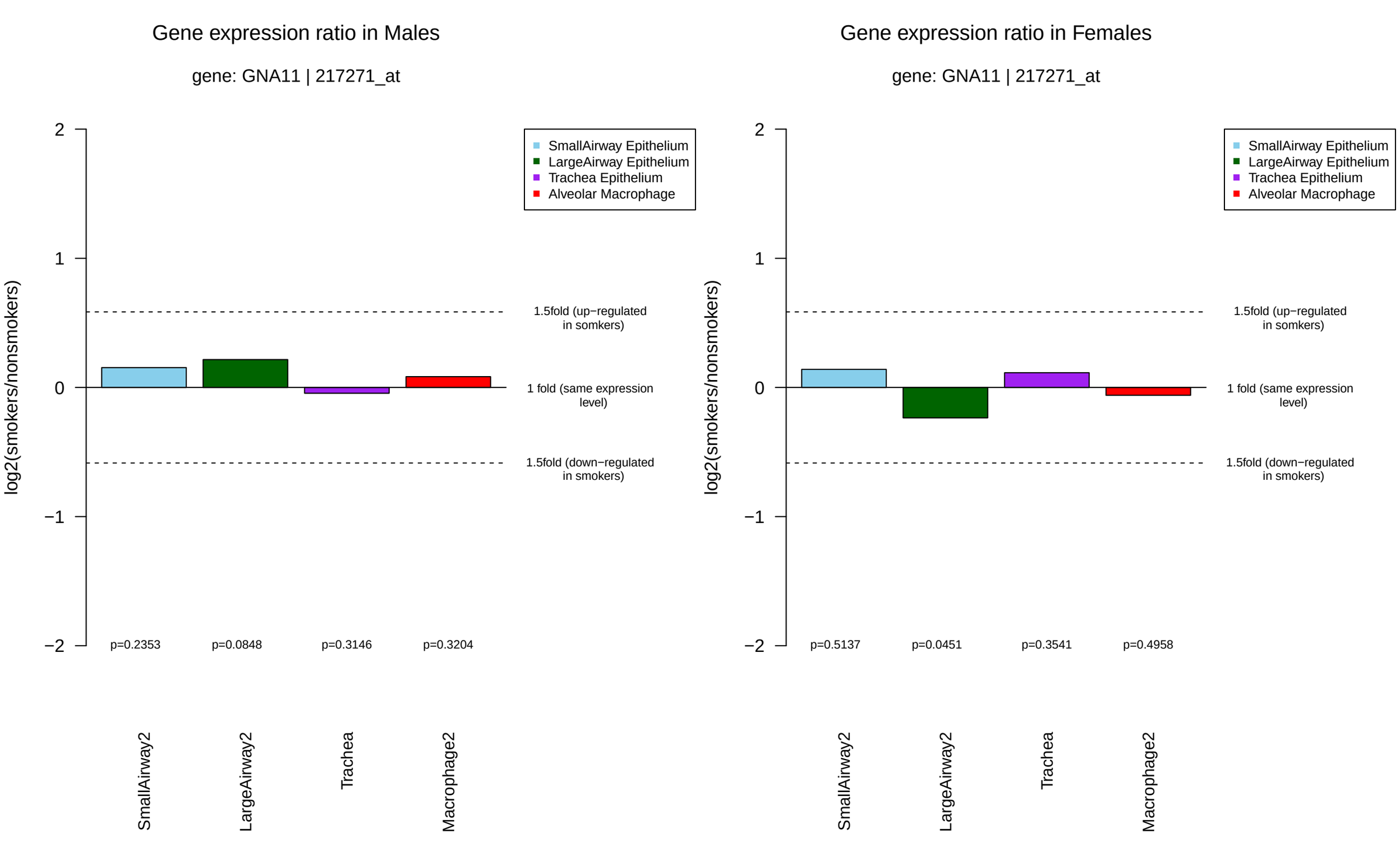
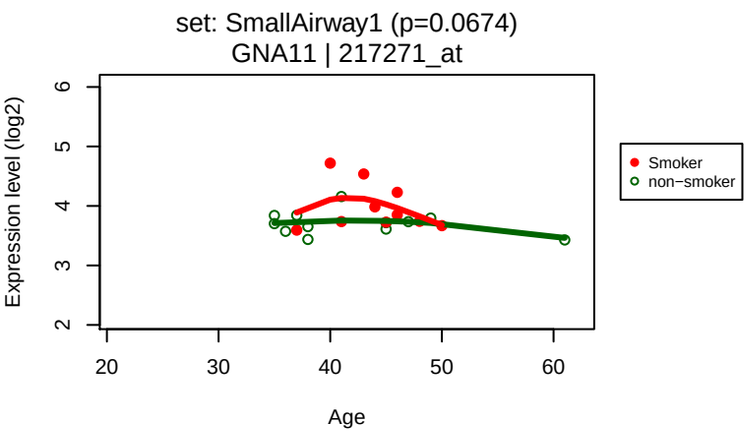
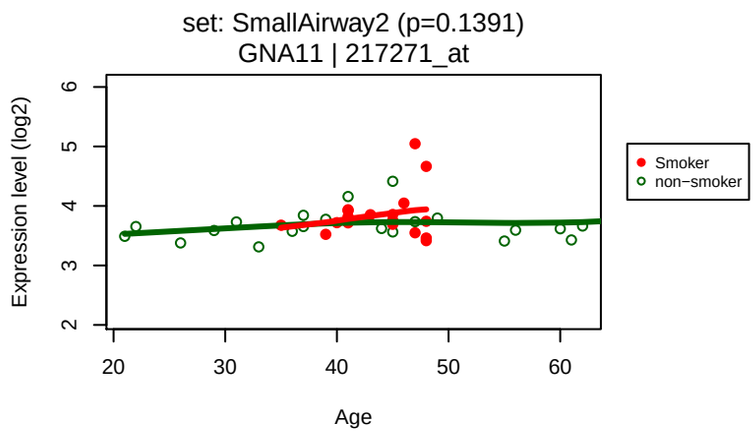
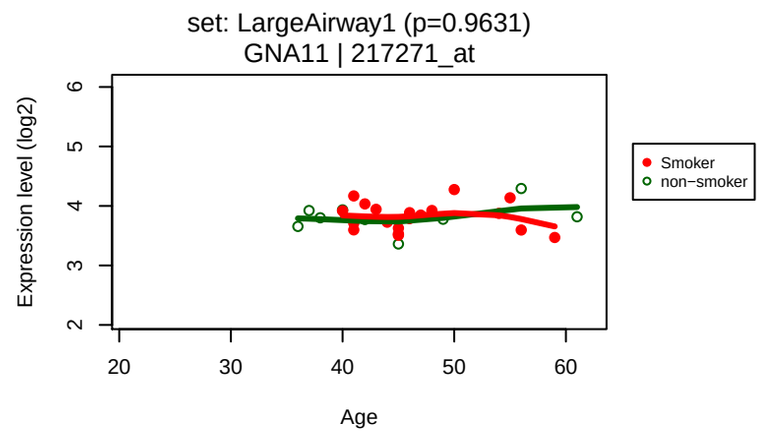

Gene expression of smokers and nonsmokers
gene: GNA11 | 217271_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0674
p=0.1391
p=0.1461
p=0.9631
p=0.4077
p=0.9753
p=0.8862
p=0.6201
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

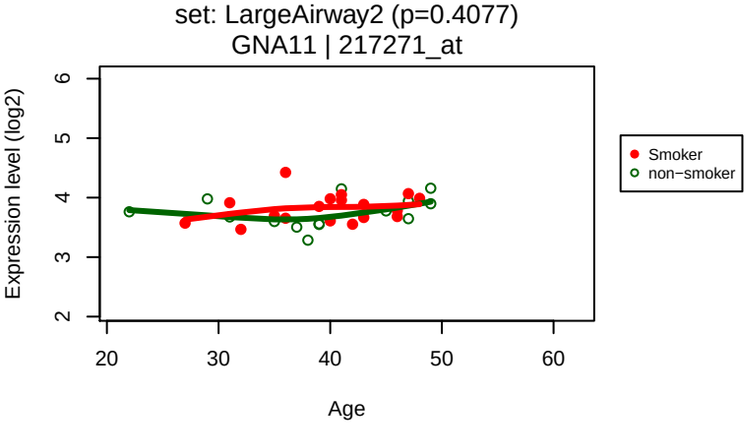

Gene expression ratio (smokers/nonsmokers)
gene: GNA11 | 217271_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0674
p=0.1391
p=0.1461
p=0.9631
p=0.4077
p=0.9753
p=0.8862
p=0.6201
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

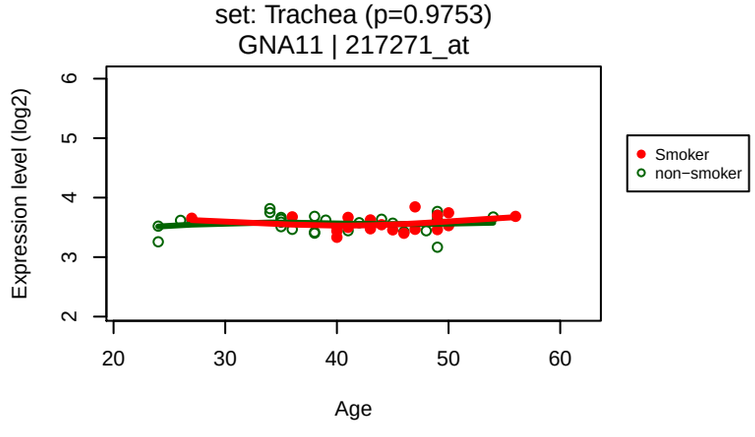

Gene expression ratio in Males
gene: GNA11 | 217271_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2353
p=0.0848
p=0.3146
p=0.3204
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GNA11 | 217271_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5137
p=0.0451
p=0.3541
p=0.4958
SmallAirway2
LargeAirway2
Trachea
Macrophage2

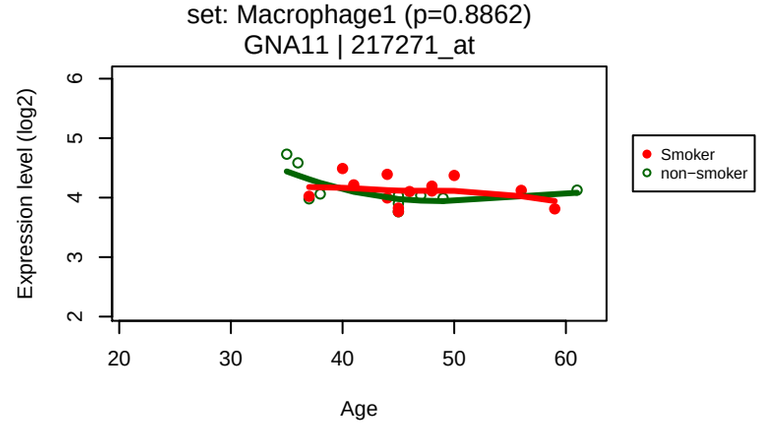

set: SmallAirway1 (p=0.0674)
GNA11 | 217271_at
Expression level (log2)
Age
Smoker
non−smoker

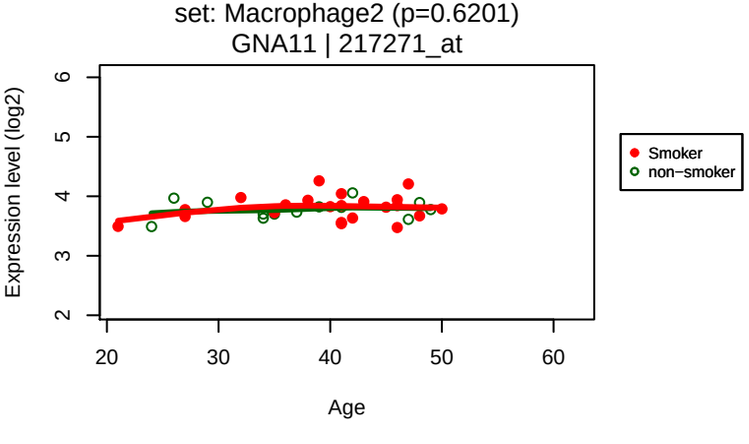

set: SmallAirway2 (p=0.1391)
GNA11 | 217271_at
Expression level (log2)
Age
Smoker
non−smoker

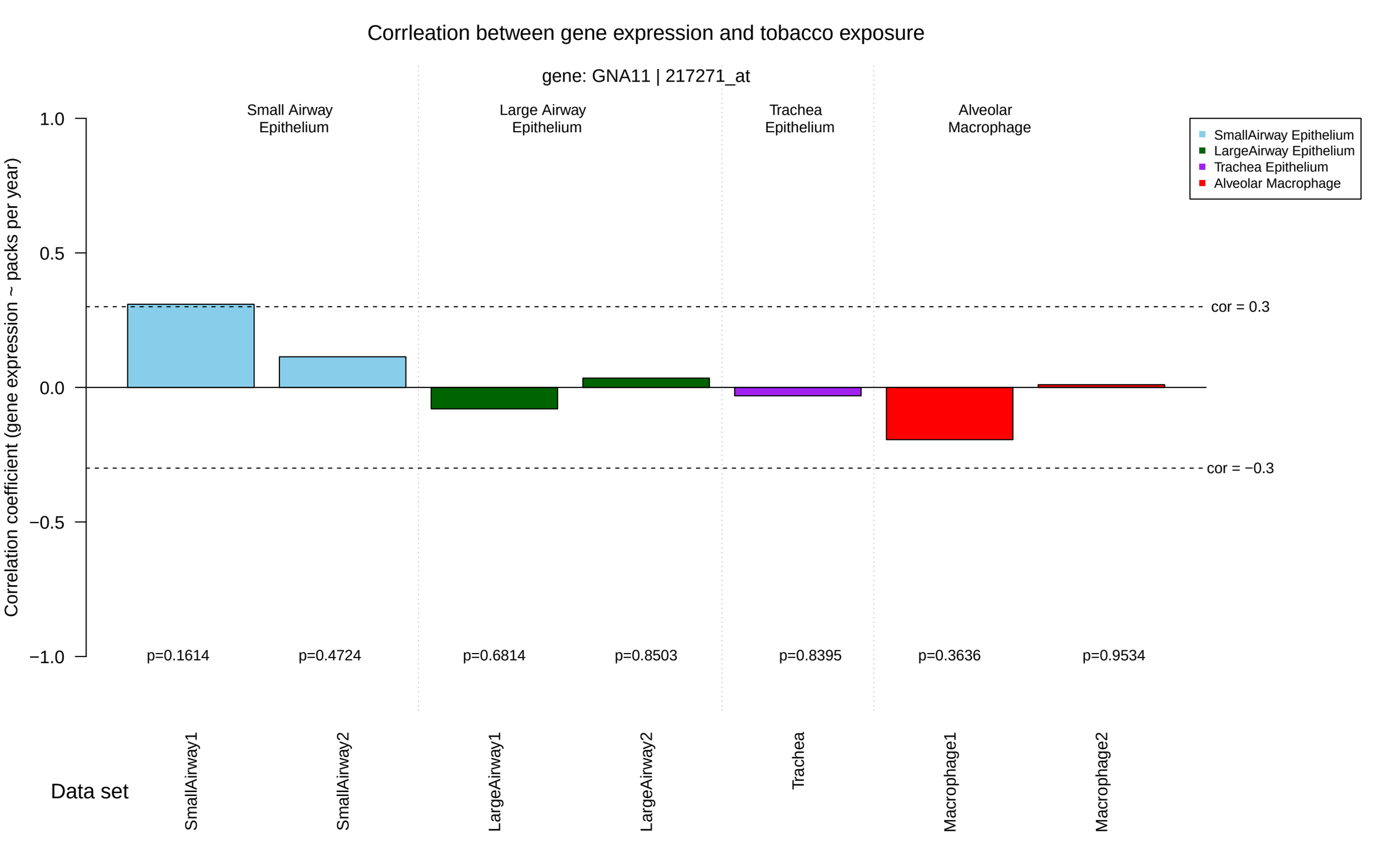

set: LargeAirway1 (p=0.9631)
GNA11 | 217271_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4077)
GNA11 | 217271_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9753)
GNA11 | 217271_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8862)
GNA11 | 217271_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6201)
GNA11 | 217271_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GNA11 | 217271_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.1614
p=0.4724
p=0.6814
p=0.8503
p=0.8395
p=0.3636
p=0.9534
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2